FOR THE COMMUNITY — HEALTH FIRST, IN PLAIN WORDS

THE ABSENCE AUDIT — THE METHOD, AT NATIONAL SCALE

Tunisia, audited — Cut the Dose

The burden measured, the cheapest cut in each chain, the 90-day pilot, the public scoreboard, and the food stack — plain words, full arithmetic.

Provenance law: MEASURED = cited anchor · DERIVED = arithmetic on measured values · ASSUMED = declared default. commercially-validated = 0 — no buyer has signed anything; published as zero on purpose. Generated 2026-09-10 by nations_audit.py / make_nations_pdfs.py — the same code that renders the page.

1. The burden, measured — and what the numbers say

Roughly **379 years of healthy life lost every year** in the basin (30-70k people): 200 to fluoride — teeth and bones — 120 to heavy metals, 59 to dust (H4-H6 anchors, DERIVED). A child in Redeyef develops asthma at age two; the nearest hospital is 70 km; the doctors' prescription is *leave Gafsa* (H16).

2. Cut the Dose — the cheapest link in each chain

CHAIN	THE CUT	COST
Fluoride: water → teeth → bones	point-of-use filters (Nalgonda/bone char)	**$2,592 per healthy year** vs the $4,000 standard — this rung pays for itself
Metals: water → gut → kidney	zinc, selenium, probiotics (gut binding)	in the $2.20M/yr full stack
Dust: air → lungs	clean-air boxes + N95-class masks (98%+ vs ≤44% cloth)	RCT: poorly-controlled asthma IRR 0.43-0.45
Fluorosis in children	calcium + vitamin C + D3 — the combination shown to REVERSE it (RCT)	pennies a day

3. The 90-day pilot — and the scoreboard that follows

$58,000: 100 households + 2 schools, 90 days, results published. **Gate:** filter uptime ≥70%, urinary F ↓20%. **The scoreboard:** water chemistry, filter uptime, cohort KPIs — public, every quarter. The health data of the basin becomes the scoreboard of the nation — and the collateral that lowers the price of every loan in the plan (100-150bp on ~$800M of portfolio debt: $8-12M/yr).

4. The food stack — protein the people can pay

PIECE	NUMBER
Spirulina ponds	10 ha → 130 t/yr → **178,082 children** at the RCT dose (2 g/day); the basin needs 0.7 ha
Protein price	$5-10/kg where imported meat protein costs $15-25/kg

PIECE	NUMBER
Terfas domestic quota	≥30% of harvest sold at home at ≤€12/kg — written into the contracts
Bone char	90 t/yr from local slaughterhouses → the filters, and a workshop of jobs
Success metric	the price of protein and vegetables in Gafsa and Gabès bends downward within 24 months of first harvest — published quarterly

5. The two futures, in plain words

Without the plan: the dose accumulates — fluoride in the teeth before age 8, cadmium in the kidneys for decades, dust in the lungs. The fittest leave: engineer to Paris, founder to London. Every departure makes the basin harder for those who stay. That future requires no decision — it is the default.

With the plan: we change the environment, not the genes. Clean water → intact enamel and strong bones within one generation. Protein → nourished brains. Jobs → a reason to stay. Same people, same geography — a different fitness landscape. The plan is evolution by other means.

What this document is: a concept-level engineering design whose numbers are computed in code from sourced primitives — recomputed, corrected, and applied into the design. Free and public.

What it is not: a commissioned state program, a bank-grade feasibility study, an investment offer, or a claim that any of this has been built. Nothing here is advice to any government or any investor. Nothing replaces geological, radiological, geotechnical or EPC-level surveys. Verify every figure and citation before relying on any of it.

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